

Collin Dang

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OVERVIEW

Recent Graduate of Computer Systems Engineering student at Carleton University with a concentration in AI/ML, ASIC Design, and Network & Software Security. Experienced across the full software development lifecycle through a co-op at Magnet Forensics and multiple engineering projects involving embedded systems, PCB design, and cloud-integrated platforms. Adept at bridging and testing software and hardware to deliver reliable, scalable solutions in fast-paced, collaborative environments.

EDUCATION

CARLETON UNIVERSITY

Bachelor of Engineering (CEAB Accredited)

Ottawa, ON

June 2026

Major in Computer Systems Engineering; Relevant Coursework: AI in Engineering, Machine Learning, Network and Software Security, Software Engineering, Operating Systems, Data Analysis

WORK EXPERIENCE

MAGNET FORENSICS (Digital Forensics and Cybersecurity Software Company)

Kanata, ON

Software Test Specialist

Jan 2024 – Aug 2024

- Developed and tested a new settings page for MAGNET AXIOM, independently programmed, manually tested, and updated automatic regression tests to ensure seamless integration and software quality.
- Collaborated with software developers and other testers to refine software and resolve issues, leading to fewer customer-reported bugs and high customer satisfaction.
- Developed new and enhanced existing C# unit tests and automated tests in Test Studio, cutting down manual testing effort and strengthening test reliability.

VIRTUCOM CABLE PRODUCTS LTD. (Cable Manufacturing Company)

Carleton Place, ON

Production Operator and Assembler

May 2023 – Aug 2023

- Optimized workstations by upgrading and configuring system hardware and software, increasing label printing speeds by 15%, and improving overall system performance for day-to-day tasks.
- Restructured an assembly line system by analyzing and reconfiguring workflows, increasing efficiency by 33% while maintaining product quality and meeting consumer expectations.
- Worked collectively with staff and engineers to ensure that constructed/modified products are to standard, which led to a smoother process, higher customer satisfaction, and a decrease in product returns.

PROJECTS

ACCESSIBLE AUTONOMOUS VEHICLE (AAV) | C++, System Design

Sept 2025 - April 2026

- Independently researched, designed, and implemented wheel tachometers for the autonomous vehicle using C++, enabling real-time speed calculations and direction detection.
- Contributed to radar, LiDAR, and GPS/GNSS systems integration, accelerating overall completion timeline through effective collaboration and technical support.
- Designed custom PCBs and 3D printed mounting components to support other teams.

SMART ATTENDANCE SYSTEM FOR STUDENTS (SASS) | Python, SQL, MFA

Jan 2025 - April 2025

- Researched and co-designed system architecture for SASS attendance platform, including RFID and QR code 2FA systems, reducing sign-in times by 15% and earning project A+ grade.
- Built and managed Firebase SQL database, enabling real-time data transfers, seamless cloud authentication, and instant connections between attendance systems and admin portals.
- Led electrical engineering efforts, hand-designing electrical systems using department hardware kits for simple, portable, scalable units.

ADDITIONAL

Technical Skills: Advanced in Python, C++, Verilog, Embedded Systems; Proficient in ASIC Design, RTOS, SQL, C/C#, MATLAB, AutoCAD, MultiSim

Training: Git, Jira, AWS, GitHub, Jupyter Notebooks, Windows, MacOS, Linux